Superpolynomial sample lower bounds for globally margin-conditioned Bayesian neural networks below the mean-field scaling

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob*

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Abstract

We study fully connected Bayesian tanh networks on the Boolean cube, conditioned before training to have a fixed absolute output margin at every input. Throughout the entire subcritical scaling regime $\gamma < 1$, we prove that $\exp(\Omega(\sqrt{k}\log k))$ examples are necessary to give the correct label probability strictly greater than $2/3$ on more than $15/16$ of the inputs of k-bit parity, in the width limit at any fixed depth. The bounds hold for every training set of the indicated size. For $\gamma \geq 0$ we also obtain a lower bound on the expected error of a posterior draw. The proof combines an explicit kernel spectral estimate with an elementary polynomial-threshold obstruction. Three distinct arguments handle the Gaussian sign posterior at $\gamma < 0$, the globally truncated Gaussian posterior at $\gamma = 0$, and the actual network's moderate-deviation minimizers at $0 < \gamma < 1$. The positive-exponent argument does not assume Gaussian posterior equivalence and allows arbitrary weights among tied rate minimizers. An $\exp(\Omega(k))$ bound at fixed positive test error is not established here.

1 Model and main result

Fix a dimension d, depth $L \geq 2$, width N, margin $a > 0$, and $\gamma < 1$. On $\mathcal{X} = \{-1, 1\}^d$, let

$$h^0(x) = (1, x), \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \ (1 \leq s < L), \qquad z_N(x) = W^L h^{L-1}(x).$$

All hidden layers have width N. The readout is linear, there are no biases after the augmented input, and the independent centered Gaussian weights have variances

$$\mathrm{Var}(W^1_{ij}) = N^{1-\gamma}/d, \quad \mathrm{Var}(W^s_{ij}) = N^{-\gamma} \ (1 < s < L), \quad \mathrm{Var}(W^L_i) = N^{-\gamma-1}. \tag{1}$$

The prior is conditioned on the all-input event $A_a = \{|z_N(x)| > a \text{ for every } x \in \mathcal{X}\}$. Training then conditions on exact, noiseless signs $\operatorname{sign} z_N(x_i) = f(x_i)$. Equivalently, the posterior is the original weight law conditioned on A_a and these observed signs. The posterior is well defined for all sufficiently large widths, as follows in particular from the positive orthant probabilities proved below; no conditioning on impossible events at smaller widths is asserted.

For a sample S, write

$$p_N(x) = \mathbb{P}\{\operatorname{sign} z_N(x) = f(x) \mid A_a, S\}, \qquad R_N(S, f) = 2^{-d} \sum_x [1 - p_N(x)],$$

*Authors are listed in alphabetical order.

and use the confidence criterion

$$F_N(S,f) = 2^{-d}\big|\{x : p_N(x) \le 2/3\}\big|. \tag{2}$$

Thus R_N is the expected error of a posterior sample, and F_N counts an input as unsuccessful unless its correct-label probability is strictly greater than $2/3$. These are different quantities; in particular this paper does not identify either with the error of the sign of the posterior latent mean.

Theorem 1.1 (A common sample lower bound for every $\gamma < 1$)**.** *Fix the depth $L \ge 2$ and margin $a > 0$, and take ambient dimension $d = k$. There are constants $c > 0$ and k_0 such that, for every $k \ge k_0$, every sample S of k-bit full parity with at most*

$$n \le \exp\big(c\sqrt{k}\log(k+1)\big)$$

distinct inputs, and every fixed $\gamma < 1$,

$$\liminf_{N\to\infty} F_N(S,f) \ge \frac{1}{16}.$$

If $0 \le \gamma < 1$, the same constants can be chosen so that $\liminf_{N\to\infty} R_N(S,f) \ge 3/8$.

The proof is completed in Section 7. The following explicit bound covers positive exponents; the more general spectral statement is Theorem 4.3.

Theorem 1.2 (A sample lower bound throughout $0 < \gamma < 1$)**.** *Let the target be full parity $f(x) = \prod_{j=1}^{k} x_j$, with $d = k$. Set*

$$c_L = (2/\pi)^{L-1}, \qquad T_k = \lceil (k+1)\log(4(k+1))\rceil + 2, \qquad r_k = \lfloor \sqrt{k}/16 \rfloor.$$

Suppose $k \ge 4096$ and

$$c_L \binom{k}{r_k+1} \ge 160(k+1), \qquad n \le \frac{c_L}{2560\,T_k}\binom{k}{r_k+1}. \tag{3}$$

For every noiseless training sample with at most n distinct inputs, and for every fixed $0 < \gamma < 1$,

$$\liminf_{N\to\infty} R_N(S,f) \ge \frac{2}{5}, \qquad \liminf_{N\to\infty} F_N(S,f) \ge \frac{1}{10}. \tag{4}$$

At each fixed depth, the sample threshold in (3) is $\exp(\Omega(\sqrt{k}\log k))$.

The sample locations can be arbitrary and can include repetitions. In particular, the conclusion holds for independent uniform training inputs without an exceptional-sample qualification. Every width limit is taken at fixed d, L, γ, a, S; the statement about growth in k concerns these limiting learners. No uniform interchange of the width and dimension limits is asserted.

Parity is a small-circuit target. Appendix A gives a bounded-weight tanh realization at every fixed depth. The lower bound therefore concerns statistical behavior of this specified prior and conditioning rule, rather than the intrinsic information needed to identify a parity function. In particular, at every fixed $\gamma < 1$ it rules out a sample guarantee polynomial in input dimension and bounded-weight network complexity, uniformly in width.

2 The kernel rate of the actual network

Let $M = 2^d$, and identify output functions with vectors in $\mathbb{R}^M$. Define

$$H(t) = \frac{2}{\pi} \arcsin t, \qquad K(x,y) = H^{\circ(L-1)}\left(\frac{1 + x \cdot y}{d+1}\right). \tag{5}$$

The positive definiteness of K is proved in the next section. For each Boolean function g, put

$$E_K(g) = \min_{g(x)z(x) \geq a \ \forall x} \frac{1}{2} z^T K^{-1} z. \tag{6}$$

The minimum exists since its feasible set is closed and nonempty and the objective is coercive. Replacing the closed inequalities by strict ones gives the same infimum, by scaling a feasible vector by $1 + \epsilon$.

Proposition 2.1 (Posterior concentration on minimum-energy signs)**.** *For the actual network* (1)*, at every fixed* $0 < \gamma < 1$*,*

$$\lim_{N \to \infty} N^{-\gamma} \log \mathbb{P}\{g(x) z_N(x) > a \ \forall x\} = -E_K(g). \tag{7}$$

Consequently, after any fixed noiseless training sample, posterior mass tends to one on the compatible functions minimizing E_K*. No assertion about the relative weights of tied minimizers is needed.*

Proof. For hidden layer s, let $Q_N^s(x,y) = N^{-1} \sum_i h_i^s(x) h_i^s(y)$. We first establish, for each fixed $\epsilon > 0$,

$$\mathbb{P}\{\|Q_N^{L-1} - K\| > \epsilon\} \leq C_\epsilon e^{-c_\epsilon N} \tag{8}$$

for all sufficiently large N. Any matrix norm can be used, since M is fixed. Conditional on the preceding layer, the rows of the next hidden layer are independent bounded vectors. Hoeffding's inequality controls all their empirical second moments. Their preactivations have covariance $N^{1-\gamma} Q_N^{s-1}$; at the first layer the covariance is $N^{1-\gamma}(1 + x \cdot y)/d$. Because $N^{1-\gamma} \to \infty$, the conditional second-moment map converges to the Gaussian-sign map

$$Q \longmapsto \left[H\left(\frac{Q(x,y)}{\sqrt{Q(x,x) Q(y,y)}} \right) \right]_{x,y}.$$

This follows by dominated convergence for a Gaussian vector, since each scalar variance stays bounded away from zero in a neighborhood of the limiting covariance. The convergence is uniform in such a neighborhood: a compactness subsequence and a continuous square root coupling reduce the assertion to the same dominated-convergence argument. The limiting map is continuous there. Inducting through the fixed number of layers, with smaller fixed tolerances at earlier layers, proves (8). The first-layer limit normalizes the diagonal $(d+1)/d$, giving the kernel in (5).

Conditionally on the hidden layers, the readout is exactly Gaussian:

$$Z_N \mid Q_N^{L-1} \sim \mathcal{N}(0, N^{-\gamma} Q_N^{L-1}). \tag{9}$$

Couple it to $Y_N = N^{-\gamma/2} K^{1/2} \xi$, where $\xi \sim \mathcal{N}(0, I_M)$ is independent of the hidden layers. On a fixed sufficiently small neighborhood of the positive-definite matrix K, the square-root map is Lipschitz. Hence, for $\delta > 0$,

$$\mathbb{P}\{\|Z_N - Y_N\| > \delta\} \leq C_\epsilon e^{-c_\epsilon N} + \mathbb{P}\{\|\xi\| > \delta N^{\gamma/2}/(C\epsilon)\}.$$

First divide logarithms by N^γ and let $N \to \infty$; then let $\epsilon \downarrow 0$. The resulting upper limit is $-\infty$, since $\gamma < 1$. The coupled vectors are therefore exponentially equivalent at speed N^γ.

The fixed-dimensional Gaussian vectors Y_N have large-deviation rate $I(z) = z^T K^{-1} z/2$, directly from their densities. Exponential equivalence transfers the same bounds to Z_N. Applying the upper bound to a closed margin orthant and the lower bound to its interior gives (7), since their infima agree. Finally, there are only finitely many Boolean functions for fixed d. Dividing each compatible orthant probability by their sum proves the posterior concentration claim. □

This is a rate statement for the network itself. It does not require relative-error approximation of rare probabilities, nor convergence in total variation to a Gaussian posterior.

Proposition 2.2 (The nonpositive-exponent posterior limits)**.** *For $\gamma = 0$, the Boolean posterior converges to that of $G \sim \mathcal{N}(0, K)$ conditioned on $|G(x)| > a$ everywhere and on the observed signs. For $\gamma < 0$, it converges to the Gaussian posterior conditioned only on the observed signs.*

Proof. The hidden-kernel argument establishing (8) holds for every $\gamma < 1$. By (9), the normalized vector $N^{\gamma/2} Z_N$ therefore converges in law to $\mathcal{N}(0, K)$. Its all-input margin is $aN^{\gamma/2}$, which equals a when $\gamma = 0$ and tends to zero when $\gamma < 0$. The limiting Gaussian is nonsingular. Thus all boundaries involved in any fixed full-sign event or margin event have probability zero, and the conditioning events have positive probability. Taking ratios of their probabilities gives convergence of each Boolean posterior mass. There are finitely many masses for fixed d. □

3 Spectral estimates with explicit dimension dependence

For $U \subseteq [d]$, let $\chi_U(x) = \prod_{j \in U} x_j$. The characters diagonalize K. Write λ_j for the eigenvalue of the normalized operator $M^{-1}K$ on degree-j characters, and let v denote the smallest eigenvalue of the matrix K itself.

Lemma 3.1 (Spectrum and minimum matrix eigenvalue)**.** *Put $D = d + 1$, $c_L = (2/\pi)^{L-1}$, and $T_d = \lceil D \log(4D) \rceil + 2$. Then*

$$\lambda_0 = \lambda_1 > \lambda_2 = \lambda_3 > \cdots > 0, \qquad v = M\lambda_d,$$

with a final unpaired value when d is even. Moreover,

$$\lambda_0 \geq \frac{c_L}{D}, \qquad v \geq \frac{c_L}{4T_d}, \qquad \lambda_j \leq \binom{d}{j}^{-1} \quad (0 \leq j \leq d). \tag{10}$$

Proof. Write $F = H^{\circ(L-1)} = \sum_{m \text{ odd}} c_m t^m$. All these coefficients are positive, sum to one, and satisfy

$$c_1 = c_L, \qquad c_m \geq c_L/m^2 \quad (m \text{ odd}). \tag{11}$$

Indeed, the coefficient of t^{2j+1} in H is $(2/\pi)\binom{2j}{j}/[4^j(2j+1)]$, and $\binom{2j}{j} \geq 4^j/(2j+1)$ because the largest of $2j+1$ binomial coefficients is at least their average. Each additional composition contributes at least its linear coefficient times the previous series.

On the full D-dimensional cube, the degree-r Fourier eigenvalue of $(u \cdot w/D)^m$ is the probability $p_m(r)$ that the simple coordinate-flip walk, started at a fixed vertex, reaches a specified vertex at distance r after m steps. Expanding the power proves this: the surviving terms are precisely those

whose coordinates selected an odd number of times are that specified support. The walk recursion yields

$$\begin{aligned} D[p_{m+1}(r) - p_{m+1}(r+2)] &= r[p_m(r-1) - p_m(r+1)] \\ &\quad + (D-r-2)[p_m(r+1) - p_m(r+3)] \end{aligned}$$

for $0 \le r \le D-2$, omitting terms with zero boundary coefficients. Starting at time zero, induction gives monotonicity with distance within each parity class. Thus the full-cube eigenvalues $\mu_r = \sum_{m \text{ odd}} c_m p_m(r)$ vanish for even r and decrease strictly through odd r: the positive term $m = r$ distinguishes r from $r+2$. Restricting the first coordinate to $+1$ gives $\lambda_r = \mu_r + \mu_{r+1}$. This proves the paired ordering and positive definiteness. The linear coefficient gives the stated lower bound on λ_0. The diagonal $K(x,x) = 1$ gives $\sum_j \binom{d}{j}\lambda_j = 1$, proving the last inequality.

To bound the smallest eigenvalue from below, let $m_0 = \lceil D \log(4D) \rceil$. The Fourier expansion of the walk shows that for $m \ge m_0$ and a vertex of matching parity, the constant and all-coordinate characters contribute $2 \cdot 2^{-D}$. The sum of absolute values of the other contributions, divided by 2^{-D}, is at most

$$\sum_{j=1}^{D-1} \binom{D}{j} |1 - 2j/D|^m \le 2[(1 + e^{-2m/D})^D - 1] < 1.$$

The last inequality follows from $De^{-2m/D} \le 1/(16D)$ and $(1+t)^D \le e^{Dt}$. Consequently $p_m(r) \ge 2^{-D}$ whenever m, r are odd and $m \ge m_0$. Using (11) and comparing a decreasing series to its integral gives, for every odd r,

$$\mu_r \ge 2^{-D} c_L \sum_{\substack{m \ge m_0 \\ m \text{ odd}}} \frac{1}{m^2} \ge \frac{2^{-D} c_L}{2(m_0+2)}.$$

Exactly one of $d, d+1$ is odd, so multiplying $\lambda_d = \mu_d + \mu_{d+1}$ by $M = 2^{D-1}$ yields $v \ge c_L/[4(m_0 + 2)]$. □

4 Low-energy outputs cannot approximate parity too well

We include the elementary polynomial-threshold argument of Aspnes, Beigel, Furst, and Rudich [1]; no threshold-complexity result is assumed beyond the proof below. At zeros a threshold function may use either fixed sign convention.

Lemma 4.1 (Polynomial thresholds and parity)**.** *If P is a real polynomial of degree at most $r < k$ on $\{-1,1\}^k$, then*

$$\mathbb{P}_x\{\operatorname{sign} P(x) \ne \chi_{[k]}(x)\} \ge b_k(r) := 2^{-k} \sum_{j=0}^{\lfloor (k-r-1)/2 \rfloor} \binom{k}{j}. \tag{12}$$

The same bound holds against a parity on k of d coordinates for a polynomial of total degree at most r on d variables. Also

$$b_k(r) \ge \frac{1}{2} - \frac{r+2}{\sqrt{2k}}. \tag{13}$$

Proof. A sufficiently small constant perturbation makes P nonzero at all cube points while preserving the chosen threshold function. Let $t = \lfloor (k-r-1)/2 \rfloor$. If its error set had fewer than $\sum_{j=0}^{t} \binom{k}{j}$ points, the linear space of multilinear polynomials of degree at most t would contain a nonzero polynomial Q vanishing on that set. Then PQ^2 is nonzero as a cube function, has degree at most

$r+2t<k$, and has nonnegative product with full parity everywhere, with a strictly positive product somewhere. Its inner product with parity is positive, contradicting Fourier orthogonality. This proves (12). For a parity on a subset of coordinates, condition on all the remaining coordinates and average the same bound.

For completeness, the largest atom of $\mathrm{Bin}(k,1/2)$ is at most $\sqrt{2/k}$. One can obtain this from $\binom{2m}{m}/4^m \le 1/\sqrt{m+1}$, proved by induction using the ratio $(2m+1)/(2m+2)$; odd k follows from the adjacent even case. By symmetry, the binomial tail in (12) differs from $1/2$ by at most $(r+2)/2$ times the largest atom. This proves (13). □

Lemma 4.2 (A cheap compatible output)**.** *For any target and any sample with at most n distinct inputs, the smallest compatible energy satisfies*

$$E_* \le a^2\left(\lambda_0^{-1}+4n/v\right). \tag{14}$$

Proof. Start with the constant vector $a\mathbf{1}$ and subtract $2a$ at each observed negative-label input. The result has absolute margin a everywhere and agrees with every observed label. Write it as $u+w$, with $u=a\mathbf{1}$ and w supported on at most n points. The quadratic norm inequality gives

$$\tfrac{1}{2}(u+w)^TK^{-1}(u+w) \le u^TK^{-1}u + w^TK^{-1}w \le a^2/\lambda_0 + 4a^2n/v.$$

This output is feasible for one compatible closed-margin orthant, which suffices for its rate energy. □

Theorem 4.3 (A spectral sample lower bound)**.** *Let $f=\chi_U$ with $|U|=k$, and let S have at most n distinct inputs. For any integer $0\le r<k$, put*

$$\eta = 2\lambda_{r+1}\left(\lambda_0^{-1}+4n/v\right).$$

Throughout $0<\gamma<1$,

$$\liminf_{N\to\infty} R_N(S,f) \ge b_k(r)-\eta, \tag{15}$$

$$\liminf_{N\to\infty} F_N(S,f) \ge \max\{0,[3b_k(r)-3\eta-1]/2\}. \tag{16}$$

In particular, with $D=d+1$, T_d as in Lemma 3.1,

$$\eta \le \frac{2[D+16nT_d]}{c_L\binom{d}{r+1}}. \tag{17}$$

Proof. Take any compatible minimizing sign function g, and let z minimize its closed-margin energy. In the normalized Fourier basis,

$$2E_* = z^TK^{-1}z = \sum_{V\subseteq[d]} \frac{\widehat{z}(V)^2}{\lambda_{|V|}}.$$

Let $P(x)=\sum_{|V|\le r}\widehat{z}(V)\chi_V(x)$. The paired ordering of the eigenvalues and Parseval's identity give

$$\mathbb{E}_x[(z(x)-P(x))^2] \le 2E_*\lambda_{r+1}.$$

Since $|z(x)|\ge a$, a disagreement between $g(x)=\operatorname{sign} z(x)$ and the chosen sign of $P(x)$ implies $|z(x)-P(x)|\ge a$. Markov's inequality and Lemma 4.2 bound the fraction of such disagreements by η. Lemma 4.1 then gives $\mathrm{err}(g,f)\ge b_k(r)-\eta$.

The estimate holds for every compatible minimizing sign, so Proposition 2.1 proves (15) irrespective of the posterior weights among minimizers. At finite width,

$$R_N \le F_N + (1 - F_N)/3 = (1 + 2F_N)/3,$$

which yields (16). Substituting the three spectral bounds from Lemma 3.1 proves (17). □

Proof of Theorem 1.2. Let $d = k$, $r = r_k$, and $B = \binom{k}{r+1}$. Conditions (3) imply

$$k + 1 + 16nT_k \le c_L B/160 + c_L B/160 = c_L B/80.$$

Thus $\eta \le 1/40$. As $k \ge 4096$,

$$b_k(r) \ge \frac{1}{2} - \frac{1/16 + 2/64}{\sqrt{2}} > \frac{17}{40}.$$

Theorem 4.3 gives R_N lower limit at least $17/40 - 1/40 = 2/5$, and F_N lower limit at least $1/10$. Finally, $r + 1 = \Theta(\sqrt{k})$ and $\binom{k}{r+1} \ge [k/(r+1)]^{r+1}$, so the bound on n is $\exp(\Omega(\sqrt{k}\log k))$ at fixed L. □

5 The Gaussian sign posterior: $\gamma < 0$

Write $M = 2^d$, let K denote the $M \times M$ limiting covariance matrix, and let $v = M\lambda_d > 0$ be its smallest matrix eigenvalue. All norms and averages on the cube below use the uniform probability measure. The RKHS norm of a vector h is $\|h\|_K^2 = h^T K^{-1} h$. Let S be any training sample, let s be its number of distinct locations, and condition $G \sim N(0, K)$ on the observed sign constraints. In the regime $\gamma < 0$ this is exactly the limiting posterior: the fixed original margin vanishes after output normalization.

Lemma 5.1 (A norm bound for the conditional mean)**.** *Let $m(x) = \mathbb{E}[G(x) \mid S]$. Then*

$$m = K_{\mathcal{X},S} K_{S,S}^{-1} m_S, \qquad \|m\|_K^2 \le s.$$

Proof. Distinct training coordinates have a nonsingular covariance matrix. Whiten their Gaussian vector as $G_S = K_{S,S}^{1/2}\xi$, with $\xi \sim N(0, I_s)$. The observed sign event is a homogeneous cone in ξ. A standard Gaussian radius is independent of its direction, so conditioning on this cone leaves $\mathbb{E}[\|\xi\|^2 \mid S] = s$. Jensen's inequality gives $m_S^T K_{S,S}^{-1} m_S \le s$. Gaussian conditioning gives the representer formula; its RKHS norm squared is exactly the preceding quadratic form. □

Lemma 5.2 (Posterior confidence gives a latent margin)**.** *Put*

$$\theta_0 = \frac{13}{20}, \qquad c_* = \sqrt{2\pi}\left(\frac{1}{e} - \frac{7}{20}\right) > 0.$$

At any unobserved input x, correct-label probability strictly larger than θ_0 implies

$$f(x)m(x) > c_*\sqrt{v}.$$

Proof. The conditioned Gaussian is log-concave, since its conditioning set is convex. Its scalar marginal $X = f(x)G(x)$ is therefore log-concave. If $\mu = \mathbb{E}X$, both its distribution function F_X and survival function are log-concave. Jensen's inequality and the probability integral transform give

$$F_X(\mu) \ge e^{-1}, \qquad \mathbb{P}(X \ge \mu) \ge e^{-1}.$$

For example, $\log F_X(\mu) \geq \mathbb{E} \log F_X(X) = -1$. Consequently $\mathbb{P}(X > 0) > \theta_0$ implies $\mu > 0$, since $\theta_0 > 1 - e^{-1}$.

Conditionally on the full vector G_S, $G(x)$ is Gaussian with variance

$$\sigma_x^2 = K(x,x) - K(x,S)K_{S,S}^{-1}K(S,x).$$

The same variance occurs in every component after conditioning only on the signs, so the scalar density is at most $(\sqrt{2\pi}\sigma_x)^{-1}$. Since $\mathbb{P}(X > \mu) \leq 1 - e^{-1}$, we obtain

$$\theta_0 < \mathbb{P}(X > 0) \leq 1 - e^{-1} + \frac{\mu}{\sqrt{2\pi}\sigma_x},$$

which implies $\mu > c_* \sigma_x$. Finally, $K \succeq vI$, and for every vector b of training coefficients,

$$\mathrm{Var}(G(x) - b^T G_S) \geq v(1 + \|b\|_2^2) \geq v.$$

Minimization over b gives $\sigma_x^2 \geq v$. □

Define the polynomial-threshold parity obstruction

$$B(d,r) = 2^{-d} \sum_{j=0}^{\lfloor (d-r-1)/2 \rfloor} \binom{d}{j} \quad (0 \leq r < d).$$

The polynomial-threshold lemma proved above says that every real polynomial of degree at most r, thresholded at zero, has error at least $B(d,r)$ against full parity. The convention at zero can be either fixed sign: adding an arbitrarily small constant removes all zeros without changing that convention.

Theorem 5.3 (A confidence bound for negative exponents)**.** *For full parity, every training sample with s distinct locations satisfies, in the $\gamma < 0$ width limit,*

$$F_{\theta_0}(S) \geq B(d,r) - \frac{s}{M} - \frac{s\lambda_{r+1}}{c_*^2 v}, \qquad 0 \leq r < d. \tag{18}$$

Here $F_{\theta_0}(S)$ is the fraction of inputs where correct-label probability is at most $\theta_0 = 13/20$. These points also fail the requested strict $2/3$ confidence criterion. Moreover, for the original finite-width posteriors,

$$\liminf_{N \to \infty} F_N(S) \geq B(d,r) - \frac{s}{M} - \frac{s\lambda_{r+1}}{c_*^2 v}.$$

Proof. Let p be the Fourier truncation of m to degree at most r. The eigenvalues decrease with Fourier degree, and Lemma 5.1 gives

$$\|m - p\|_2^2 \leq \lambda_{r+1} \|m\|_K^2 \leq s\lambda_{r+1}.$$

At an unobserved input with confidence strictly larger than θ_0, Lemma 5.2 gives $f(x)m(x) > c_*\sqrt{v}$. If the sign of $p(x)$ is wrong there, then $|m(x) - p(x)| \geq c_*\sqrt{v}$. Thus the error of the polynomial threshold p is at most $F_{\theta_0}(S) + s/M + s\lambda_{r+1}/(c_*^2 v)$. The polynomial-threshold lower bound proves the claim. The term s/M is needed because observed coordinates do not have a positive Gaussian residual variance. □

6 A fixed global margin: $\gamma = 0$

Now the limiting posterior is $G \sim N(0, K)$ conditioned on $|G(x)| > a$ at every input and on the observed labels. This conditioning is not convex. The argument in the previous section therefore does not apply. Instead we control the Fourier tail of an entire posterior draw.

Lemma 6.1 (High-degree covariance mass)**.** *Put $\beta = 2^{-(L-1)}$ and let $\tau_r = \sum_{j>r} \binom{d}{j} \lambda_j$. For every $r \geq 0$,*

$$\tau_r \leq \frac{2}{1 - e^{-1}} (r+1)^{-\beta}.$$

Proof. Write the saturated kernel function as $F(t) = H^{\circ(L-1)}(t) = \sum_{m \geq 0} c_m t^m$, where $c_m \geq 0$ and $\sum_m c_m = 1$. The kernel $((1 + x \cdot y)/(d+1))^m$ has Fourier degree at most m and normalized trace one. Consequently $\tau_r \leq \sum_{m>r} c_m$. For $u = 1/(r+1)$,

$$(1 - e^{-1}) \sum_{m>r} c_m \leq 1 - F(1-u).$$

The elementary inequality $1 - H(1-u) \leq \sqrt{2u}$, valid for $0 \leq u \leq 1$, follows from $u = 2\sin^2(\frac{1}{2}\arccos(1-u))$ and $\sin t \geq 2t/\pi$ on $[0, \pi/2]$. Iteration gives $1 - F(1-u) \leq 2u^\beta$. □

Lemma 6.2 (A denominator bound uniform in the labels)**.** *Set*

$$t = \sqrt{2\log(4M)}, \qquad b = a + t + 1, \qquad D_s = b^2 \left(\lambda_0^{-1} + \frac{4s}{v} \right).$$

The event A_S imposing the global margin and any consistent labels on s distinct inputs satisfies

$$\mathbb{P}(A_S) \geq \tfrac{1}{2} e^{-D_s}.$$

Proof. Since every marginal of G has variance one, a union bound gives $\mathbb{P}(E) \geq 1/2$ for the symmetric set $E = \{\max_x |G(x)| \leq t\}$. Let u be the indicator vector of the observed locations with negative labels, and put $h = b(\mathbf{1} - 2u)$. Then $h + E \subset A_S$. The constant vector has squared RKHS norm $1/\lambda_0$, while $\|u\|_K^2 \leq s/v$. Thus

$$\tfrac{1}{2}\|h\|_K^2 \leq b^2(\lambda_0^{-1} + 4s/v) = D_s.$$

The Gaussian shift identity and symmetry of E give

$$\begin{aligned} \mathbb{P}(G \in h + E) &= e^{-\|h\|_K^2/2} \mathbb{E}[\mathbf{1}_E(G) e^{-h^T K^{-1} G}] \\ &\geq e^{-\|h\|_K^2/2} \mathbb{P}(E). \end{aligned}$$

Indeed the conditional mean of the linear form on the symmetric set E is zero, so Jensen's inequality applies. This proves the bound. □

Theorem 6.3 (A confidence bound at zero exponent)**.** *Fix $0 \leq r < d$ and $\delta > 0$ with $\tau_r \leq \delta/4$. Put $\ell = \lambda_{r+1}$ and*

$$q_s = \min\{1,\ 2\exp(D_s - \delta/(8\ell))\}.$$

For full parity, the limiting posterior mean error $R(S)$ and confidence failure $F(S)$ satisfy

$$R(S) \geq B(d, r) - \delta/a^2 - q_s, \tag{19}$$

$$F(S) \geq \max\left\{0, \frac{3[B(d,r) - \delta/a^2 - q_s] - 1}{2}\right\}. \tag{20}$$

Proof. Let $G_{>r}$ be the high-degree Fourier part of the Gaussian draw. In an orthonormal Fourier basis, $W = \|G_{>r}\|_2^2 = \sum_{|U|>r} \lambda_{|U|}\xi_U^2$, with independent standard Gaussian ξ_U. Every coefficient is at most ℓ, and their sum is τ_r. For $\theta = 1/(4\ell)$,

$$\log \mathbb{E}e^{\theta W} = -\tfrac{1}{2}\sum_{|U|>r} \log(1 - 2\theta\lambda_{|U|}) \le 2\theta\tau_r.$$

Markov's inequality and $\tau_r \le \delta/4$ imply

$$\mathbb{P}(W > \delta) \le e^{-\delta/(8\ell)}.$$

Lemma 6.2 therefore gives $\mathbb{P}(W > \delta \mid A_S) \le q_s$.

On A_S, if $\operatorname{sign} G$ and the threshold of $G_{\le r}$ disagree at an input, then $|G_{>r}(x)| > a$. When $W \le \delta$, their disagreement fraction is consequently at most δ/a^2. The degree-r polynomial threshold has parity error at least $B(d, r)$. Hence the posterior draw has error at least $B(d, r) - \delta/a^2$ except on an event of probability at most q_s. Averaging gives (19). Finally, if $p(x)$ is correct-label posterior probability, then $R = \mathbb{E}_x(1 - p(x)) \le (1 - F)/3 + F$. Rearranging proves (20). □

6.1 Consequence at fixed depth

For fixed L and fixed margin $a > 0$, the preceding bounds imply constants $c > 0$ and d_0 such that, for $d \ge d_0$, every sample with

$$n \le \exp(c\sqrt{d}\log(d + 1))$$

has limiting confidence failure at least $1/16$, for every $\gamma \le 0$. To see this, take $r = \lfloor\sqrt{d}/64\rfloor$. Elementary central-binomial estimates give $B(d, r) \ge 7/16$ for all sufficiently large d. Spectral ordering and trace one give $\lambda_{r+1} \le \binom{d}{r+1}^{-1} = \exp[-\Omega(\sqrt{d}\log(d + 1))]$. The spectral bounds proved earlier give $\lambda_0^{-1} = O_L(d)$ and $v^{-1} = O_L(d\log(d + 1))$. For negative exponents these estimates apply directly to (18); its two error terms tend to zero when c is sufficiently small. At exponent zero choose $\delta = a^2/32$. Lemma 6.1 eventually gives $\tau_r \le \delta/4$, and the same estimates ensure $D_n \le \delta/(16\ell)$ and $q_n \le 1/32$. Thus $R \ge 3/8$ and $F \ge 1/16$. The threshold d_0 can be very large at large fixed depth because the trace exponent is $2^{-(L-1)}$.

These are bounds for the limiting posterior, and they also give $\liminf_{N\to\infty} F_N(S) \ge 1/16$ for the requested finite-width strict $2/3$ criterion. At negative exponents, the proof uses $\theta_0 = 13/20 < 2/3$: every input counted by limiting F_{θ_0} remains a failure at all sufficiently large widths. The input cube is finite, so this holds simultaneously on the counted inputs. At exponent zero, the risk lower bound is continuous under finite-state posterior convergence; the universal inequality $F_N \ge (3R_N - 1)/2$ transfers it directly.

7 The common conclusion

Proof of Theorem 1.1. For $0 < \gamma < 1$, Theorem 1.2 supplies an explicit sample threshold of order $\exp(\Omega(\sqrt{k}\log k))$, with risk lower bound $2/5$ and confidence-failure lower bound $1/10$. For $\gamma \le 0$, the fixed-depth consequence in the preceding section gives the confidence bound $1/16$; at $\gamma = 0$ its risk bound is $3/8$. Decrease the constant c and enlarge k_0 to satisfy the finitely many displayed sufficient inequalities. Proposition 2.2, together with the strict threshold slack used for negative exponents, transfers the nonpositive statements to the claimed finite-width lower limits. □

8 What the bound does and does not establish

The result establishes a fixed-positive-error obstruction for the actual globally conditioned Bayesian learner. In particular, it is stronger than a statement that the exact parity function has tiny prior mass or cannot be the most likely exact function before every point is observed. The confidence conclusion concerns the marginal label probabilities in (2).

The exponential-in-k sample question remains open. The present method first upper-bounds the energy of the learner's minimizers and then bounds what any function of that energy can accomplish. Its limitation is concrete: a polynomial with roots between consecutive Hamming levels in a central band of width $O(\sqrt{k})$ has degree $O(\sqrt{k})$, and its sign agrees with parity on a fixed large fraction of the cube. Scaling that polynomial to a global margin gives kernel energy $\exp(O(\sqrt{k}\log k))$. Thus energy alone cannot give an $\exp(\Omega(k))$ necessary scale at fixed error. Here is the spectral justification for that energy estimate. For $j \le r$, choose the odd integer $m \in \{j, j+1\}$. The coefficient estimate (11) and the walk event which flips each of m specified coordinates once give

$$\lambda_j \ge \frac{c_L m!}{m^2(k+1)^m} \ge \frac{c_L}{(r+1)^2(k+1)^{r+1}}.$$

Using integer roots of parity opposite to k in the variable $\sum_i x_i$ makes the central-band polynomial P satisfy $1 \le |P(x)| \le (2k)^r$ everywhere. Its Fourier degree is at most $r = O(\sqrt{k})$. The feasible output aP therefore has energy at most $[a^2/(2c_L)](r+1)^2(k+1)^{r+1}(2k)^{2r}$, as claimed. A stronger sample lower bound would have to use more about the compatible minimum-energy interpolants and the training constraints. Conversely, that approximation does not prove a learning upper bound: its wrong-sign outputs outside the central band can be very large, so correcting observations there need not be cheap.

The three exponent regimes require different arguments. In particular, convex conditioning of the Gaussian sign posterior is used only for negative exponents. Neither global absolute-margin conditioning nor the positive-exponent energy minimization is a convex sign constraint.

A A small bounded-weight parity network

For completeness, full parity on k bits has a one-hidden-layer tanh realization of width $k+1$, with weights $O(k\log(k+1))$ and output margin greater than $1/2$. Let

$$t(x) = \frac{k + \sum_i x_i}{2} \in \{0, \dots, k\}, \quad y_j = (-1)^{k-j}, \quad A = \log(16(k+1)).$$

The step functions $s_j(t) = \operatorname{sign}(t - j + 1/2)$ satisfy

$$y_t = c + \sum_{j=1}^{k} y_j s_j(t), \qquad c = (y_0 + y_k)/2 \in \{0, 1\}.$$

Replace each step by $h_j = \tanh(A(t - j + 1/2))$. Since its argument has absolute value at least $A/2$, its error is at most $2e^{-A}$. A constant neuron $h_0 = \tanh 1$ supplies the term c using output weight $c/\tanh 1$. The resulting output differs from parity by less than $1/8$. First-layer input weights are $A/2$ and bias weights have magnitude at most $Ak/2$; readout weights have magnitude at most 2.

At depth $L \ge 3$, the next hidden layer sums these features and applies tanh, producing the correct sign with magnitude at least $\tanh(7/8) > 1/2$. Each additional hidden layer can preserve this margin by a single weight 2, since $\tanh(2u) > 1/2$ whenever $u > 1/2$. A final readout weight 1 completes the construction. Unused neurons are set to zero. Thus the same polynomial bounds hold at every fixed depth.

References

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